tail, recent interactions may overshadow long-term preferences and inject noise [144]. In collaborative settings, this challenge intensifies: selecting exemplar users whose preferences are genuinely similar requires careful similarity computation to avoid amplifying biases from noisy neighborhoods [187]. Cross-lingual aliasing and transliteration are also a common issue; if all seeds belong to a single language, the model may amplify that linguistic or regional bias unless the selection is balanced [42]. Moreover, session-based recommendation work documents strong order sensitivity; ICL inherits this hazard unless the selected examples are stress-tested with shuffle, drop, or paraphrase perturbations [96, 120].

Beyond personalization, ICL also opens new design levers for controllability. LLMs can interpret heterogeneous seeds—track IDs, textual notes (e.g., "no harsh vocals"), or micro-histories—and synthesize them with open-vocabulary constraints, effectively producing a local curation policy "in context" [13]. Designers can encode fairness or exploration directly (e.g., balancing genders or regions in exemplar artists), a controllability lever that classical embedding-only CF did not expose [173]. Because seed examples exist exclusively as textual tokens within the prompt context, practitioners can perform A/B testing on prompt templates, system roles [30], and instructions without requiring model retraining or parameter updates [13].

Retrieval-Augmented Generation (RAG) (user request and retrieved documents, possibly conditioned by user profile | no in-context examples or reasoning scaffold). RAG enriches the prompt with retrieved documents drawn from either the catalog or an external knowledge base (e.g., item metadata records, artist profiles, release information, user reviews, social tags, or new-release announcements) [84]. Evidence retrieved from the catalog enables the model to cite specific release groups and track identifiers, which can subsequently be mapped back to the catalog for execution. The fundamental goal is to ground recommendations in factual evidence and mitigate hallucinations by combining parametric knowledge encoded in model weights with non-parametric access to up-to-date external information [55]. This paradigm is increasingly adopted to ensure catalog freshness by leveraging current information from external sources, since new releases and reissues appear continuously in music streaming platforms [39].

Music catalogs present unique retrieval challenges that distinguish them from general-domain applications [136]. The same song may exist in multiple versions—original recordings, remastered editions, live performances—and metadata is frequently inconsistent or multilingual across databases. To address such complexities, RAG systems might employ two complementary search strategies. First, exact matching through catalog identifiers and structural relationships efficiently retrieves known items with high precision. Second, semantic search—which compares the contextual meaning of text rather than surface-level keyword matching—discovers relevant music even when users describe requests informally or in non-English languages [174]. The system subsequently validates that retrieved items exist in the catalog and resolves ambiguities, such as disambiguating between multiple artists sharing identical names or identifying cover versions among variant recordings.

Two practical considerations fundamentally shape RAG implementation in music recommendation systems. First, computational and economic efficiency: including excessive retrieved content in the prompt increases both token consumption and processing latency, requiring careful ranking and selection of only the most relevant information to maintain real-time response guarantees. Second, catalog freshness and dynamic updates: because the retrieval index resides external to the LLM's parametric weights, music streaming services can immediately add newly released tracks to search indices without model retraining or fine-tuning [84]. This decoupled architecture enables LLMs to effectively bridge informal user requests (e.g., "upbeat workout music") with formal catalog metadata (genre classifications, tempo measurements, energy ratings) [39]. When appropriately validated through entity linking and relevance filtering,

this approach enables scalable, factually grounded recommendations across millions of tracks while maintaining responsiveness to catalog evolution [55].

2.3.1 Chain-of-Thought / Tree-of-Thought (CoT/ToT) (user request and reasoning scaffold, possibly conditioned by user profile | no retrieved documents or in-context examples). Reasoning scaffolds guide LLMs to decompose complex recommendation tasks into explicit intermediate steps. For example, a CoT prompt for creating a workout playlist might instruct: *"Generate a 30-minute workout playlist by: (1) selecting high-energy tracks with BPM 140-160, (2) filtering for motivational lyrics, (3) ordering by increasing intensity, (4) ensuring smooth key transitions between adjacent tracks, and (5) justifying each transition."* Rather than producing recommendations through a single opaque generation pass, the model articulates each decision stage, making the curation process transparent and auditable.

Chain-of-Thought (CoT) structures reasoning as a linear sequence of decisions (e.g., "filter by mood → constrain by era → match key/BPM → order for flow → justify transitions"), where each step's output feeds into subsequent stages [172]. Tree-of-Thought (ToT) extends this by exploring multiple reasoning branches in parallel and aggregating across paths through self-consistency mechanisms or voting, improving robustness when single-path reasoning proves brittle [165, 176]. Both approaches enable the model to reason explicitly over multiple constraints, candidate selections, ordering criteria, and transition justifications.

Applied to music recommendation, these scaffolds provide three key benefits. First, *interpretability*: users and curators can inspect why each track was selected and verify reasoning against catalog metadata. Second, *controllability*: policy constraints (e.g., regional quotas, content restrictions, fairness objectives) can be encoded directly into the reasoning instructions without model retraining. Third, *multi-constraint handling*: the system can balance competing musical dimensions (mood, tempo, key, energy progression) through explicit deliberation rather than implicit optimization.

However, music-specific challenges require careful handling. Subjective tags (mood, energy) are culturally dependent and annotator-specific, and without explicit grounding in catalog data, models may construct plausible-sounding reasoning chains that rely on invented micro-genres or spurious correlations [71]. Even objective metadata (key, tempo) can vary across different recordings or remasters of the same composition. Furthermore, effective playlist generation requires global sequence-level coherence (smooth transitions, narrative arc, energy flow) beyond individual track quality, necessitating evaluation at pairwise and playlist-level granularity [136].

Critically, reasoning scaffolds should remain decision-useful, meaningfully improving selection quality and constraint adherence rather than producing verbose but uninformative justifications. When properly grounded in catalog data and validated against factual metadata, CoT/ToT enables scalable, interpretable music curation that supports complex multi-stage reasoning while maintaining transparency and controllability.

3 LLMs in Music Recommendation: Evaluation

In the previous section, we discussed the challenges associated with user modeling, item modeling, and natural language–based recommendation in the music domain. We also summarized the key changes introduced by the adoption of LLMs in each of these areas. In what follows, we shift our focus to evaluating LLMs when used in any of the previously discussed scenarios.

We begin by providing a comprehensive overview of how LLMs are evaluated for generative tasks, as designed and discussed within the NLP community (Section 3.1). We also outline several research gaps and methodological challenges that arise when adapting these evaluation approaches, in particular to the MRS domain. In the second part, we extend the discussion to MRS-specific evaluation (Section 3.2), first addressing success and risks dimensions, then

examining other open issues related to item and user modeling. The goal of this section is pedagogical but also critical by highlighting the main challenges and research opportunities regarding the evaluation of LLMs when used for MRS.

3.1 NLP-Driven Evaluation

When using LLMs for user modeling, item modeling, or music recommendation, the task generally reduces to learning $P_\theta(y \mid x)$, a conditional probability distribution over output texts y (e.g., user preference profiles, playlist captions, music recommendations in natural language), given a textual or multimodal input x (e.g. track metadata, text-based listening history, tokenized audio etc.). For simplicity, the task can also be expressed as an input-to-text mapping function as follows:

$$f : X \to Y, \quad x \in X,\ y \in Y,$$

where X is the set of possible inputs and Y is the set of possible output texts.

Evaluation strategies can focus either on comparing the predicted output y against a reference output y^* or human judgments (**reference-based evaluation**), or on comparing the predicted output y directly against the source input x (**reference-free evaluation**). Reference-based evaluation, which relies on ground-truth data, remains the most common approach. However, reference-free evaluation based on LLMs has recently emerged as a promising, resource-efficient alternative, as it does not require extensive human annotation and is therefore suitable for fast and large-scale iterations.

In what follows, we set aside aspects related to content parsing. Modern LLMs can generate structured outputs directly (e.g., JSON files), which provides more suitable formatting of results for evaluation. Therefore, we do not consider evaluation metrics related to this parsing phase and focus only on the quality of the generated content itself.

3.1.1 Reference-based Evaluation. For the previously described roles of LLMs, the evaluation of success when references are available can be discussed as:

(1) **Exact Match Evaluation**, used when the output can be represented in a well-defined, discrete space with unambiguous ground truth (i.e., limited number of outcomes or outcomes expected to be limited to parts of input in case of information extraction). This includes task such as NER (i.e., matching predicted music entities to reference spans, either token-wise or entity-wise) [40, 61], emotion annotation (e.g., music classification into multiple discrete classes [175]), and knowledge base linking (e.g., checking whether a predicted item identifier corresponds exactly to the reference entity in the knowledge base [119]). Formally, given a ground-truth output y^* and a model-generated output $\hat{y} = f(x)$, the **evaluation metric** can then be defined as a function:

$$M : Y^* \times Y \to \mathbb{R}, \quad Y^* \subseteq Y \quad \text{and} \quad |Y^*| \ll |Y|$$

which measures the degree of similarity or correspondence between the reference and predicted outputs:

$$M(y^*, \hat{y}) \in \mathbb{R}.$$

As Y^* consists of a limited number of outcomes (e.g., a set of class labels, named entities, or knowledge base identifiers), M corresponds to exact-match success metrics such as **Accuracy**, **Precision**, **Recall**, or **F1-score**. Ranking metrics such as **Precision@k**, **Candidate Recall Rate** could be also used, for instance, when assessing the linking of an entity to a knowledge base [131].

(2) **Approximate Match Evaluation**, used when the output is free-form text or belongs to a large, open vocabulary where multiple correct answers exist. Examples include music text summarization [179], music captioning [53, 58], or user profile generation. Formally, when Y is the output space of open-form textual possibilities (i.e.,

any natural language text), the evaluation metric M can be defined as a quality estimation function between the reference and predicted texts:

$$M : Y \times Y \to \mathbb{R}, \quad M(y^*, \hat{y}) = S(\phi(y^*), \phi(\hat{y})),$$

where $\phi(\cdot)$ denotes a transformation (e.g., embedding model, sparse feature extractor) that maps text into a continuous representation, and $S(\cdot, \cdot)$ is either a similarity measure based on distance alignment, frequency-representation of n-grams overlap, or the probability of $\hat{y}$ being generated from y^* when estimated with another pre-trained generative model. In this setting, higher $M(y^*, \hat{y})$ values indicate closer semantic alignment between the reference and generated texts. Typical choices of M for this type of evaluation include:
→ metrics based on n-gram overlap (**BLEU** [112]— precision-based metric, **ROUGE** [88]—recall-based metric, **METEOR** [106] and **CIDEr** [160]—F-measure-based metrics),
→ metrics based on semantic similarity (**BERTScore** [188], **MoverScore** [189], **BaryScore** [26], **DiscoScore** [190]),
→ supervised metrics (i.e., regressive models on top of text encoder to predict human judgements—**BLEURT** [142]).
→ metrics based on text generation (e.g. **BARTScore** [180]))

An overview of these evaluation metrics, exemplified when used for item modeling, is presented in Table 2.

Although widely adopted in the community, the approximate match evaluation metrics have been increasingly scrutinized. Studies have shown that n-gram–based metrics could correlate poorly with human judgments [99, 117], while similarity-based metrics can be overly sensitive to adversarial perturbations or lexical variations [133]. To address these limitations, a new line of research [22, 69] has proposed designing evaluation metrics M inspired by the **Natural Language Inference** (NLI) paradigm, where a model-generated output $\hat{y}$ is considered high quality if it is semantically equivalent to a human reference y^* under the notion of *bi-implication* (i.e., logical equivalence). Furthermore, combining traditional metrics with NLI- or QA- (Question Answering) based approaches has been shown to yield more robust and reliable evaluation signals, improving correlation with human assessments [22, 60].

Task Type	**Example Output**	**Common Metrics**
Classification	"high valence" / "low valence"	Accuracy, Precision, Recall, F1, ROC-AUC
Extraction (NER)	track=Love, artist=The Beatles	Precision, Recall, F1-score (token- or entity-level), Exact Match, Approximate Match
Entity Linking	Knowledge based identifier: `track:12345`	Accuracy, Precision, Recall, F1, Candidate Recall Rate, Precision@k
Summarization	"A warm, soulful album blending jazz and funk."	ROUGE, BLEU, METEOR, BERTScore, Human judgments (fluency, informativeness)
Captioning	"A mellow jazz ballad with soft piano and vocals."	BLEU, ROUGE, CIDEr, BERTScore, Human evaluation (naturalness, relevance)

Table 2. Evaluation metrics for various item data augmentation tasks achieved with LLMs.

Overall, the evaluation of natural language generation text with the **reference-based framework** remains an active and evolving area of research. *Current findings underscore the need to develop automatic success metrics that align more closely with human judgments, particularly across complex and multi-dimensional aspects of text quality. Moreover, there has been relatively little investigation into how general-domain evaluation metrics perform in specialized contexts, such as*